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Calibration results
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Normalized Residuals
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Reprojection error (cam0):     mean 0.41290037906022403, median 0.3720962058376376, std: 0.23572066968282723
Reprojection error (cam1):     mean 0.35816427023297165, median 0.3422427965292065, std: 0.17860600003967966
Gyroscope error (imu0):        mean 17.005033796026726, median 13.057021746314362, std: 13.198381248633197
Accelerometer error (imu0):    mean 0.1421181959861128, median 0.11077252108149674, std: 0.1163942474408821

Residuals
----------------------------
Reprojection error (cam0) [px]:     mean 0.41290037906022403, median 0.3720962058376376, std:
0.23572066968282723
Reprojection error (cam1) [px]:     mean 0.35816427023297165, median 0.3422427965292065, std:
0.17860600003967966
Gyroscope error (imu0) [rad/s]:     mean 145.93164831117264, median 112.05109782961385, std: 113.26419892810473
Accelerometer error (imu0) [m/s^2]: mean 0.47772568501266977, median 0.37235822019165965, std:
0.3912554701697947

Transformation (cam0):
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T_ci:  (imu0 to cam0):
[[ 0.00362718 -0.99996398 -0.00767311  0.07143705]
 [-0.00700282  0.00764757 -0.99994624  0.01980756]
 [ 0.9999689   0.00368072 -0.00697483 -0.01729721]
 [ 0.          0.          0.          1.        ]]

T_ic:  (cam0 to imu0):
[[ 0.00362718 -0.00700282  0.9999689   0.01717627]
 [-0.99996398  0.00764757  0.00368072  0.07134666]
 [-0.00767311 -0.99994624 -0.00697483  0.02023399]
 [ 0.          0.          0.          1.        ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
0.006975568193872428


Transformation (cam1):
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T_ci:  (imu0 to cam1):
[[ 0.017795   -0.99977544 -0.01150684 -0.17850725]
 [-0.0006858   0.01149645 -0.99993368  0.01920904]
 [ 0.99984142  0.01780171 -0.00048107 -0.01114729]
 [ 0.          0.          0.          1.        ]]

T_ic:  (cam1 to imu0):
[[ 0.017795   -0.0006858   0.99984142  0.01433523]
 [-0.99977544  0.01149645  0.01780171 -0.17848956]
 [-0.01150684 -0.99993368 -0.00048107  0.01714835]
 [ 0.          0.          0.          1.        ]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
0.005409889749631557

Baselines:
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Baseline (cam0 to cam1):
[[ 0.99989227  0.00373575  0.01419481 -0.24976508]
 [-0.00382593  0.99997264  0.00633091 -0.00021516]
 [-0.01417077 -0.00638453  0.99987921  0.00728661]
 [ 0.          0.          0.          1.        ]]
baseline norm:  0.24987143477713597 [m]


Gravity vector in target coords: [m/s^2]
[-0.17018629 -9.73549714 -1.16599952]


Calibration configuration
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cam0
-----
```

Camera model: pinhole
Focal length: [1746.4539465334938, 1744.4669421726232]
Principal point: [958.9950621906956, 562.5906820013532]
Distortion model: radtan
Distortion coefficients: [-0.25701113919362983, 0.10262664411100493, 0.00043516222338876745, -2.646838006691349e-05]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.088 [m]
Spacing 0.026399999999999996 [m]

cam1

Camera model: pinhole
Focal length: [1744.9386686543533, 1742.546552845488]
Principal point: [939.6720202553483, 569.8432138862356]
Distortion model: radtan
Distortion coefficients: [-0.25645546865320074, 0.1025772034872798, -0.0003740545335494358, -0.00022666394863838954]
Type: aprilgrid
Tags:
Rows: 6
Cols: 6
Size: 0.088 [m]
Spacing 0.026399999999999996 [m]

IMU configuration
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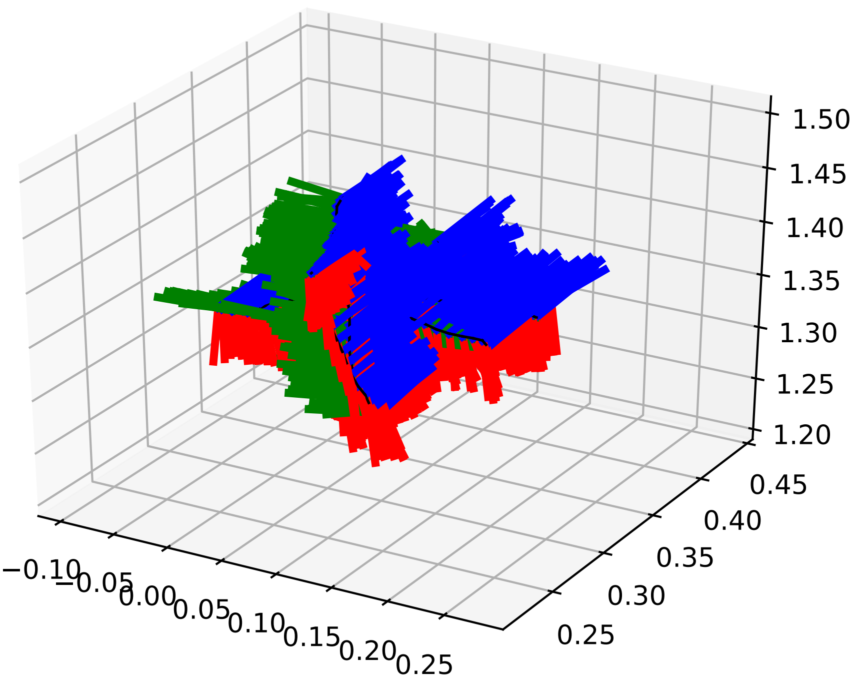
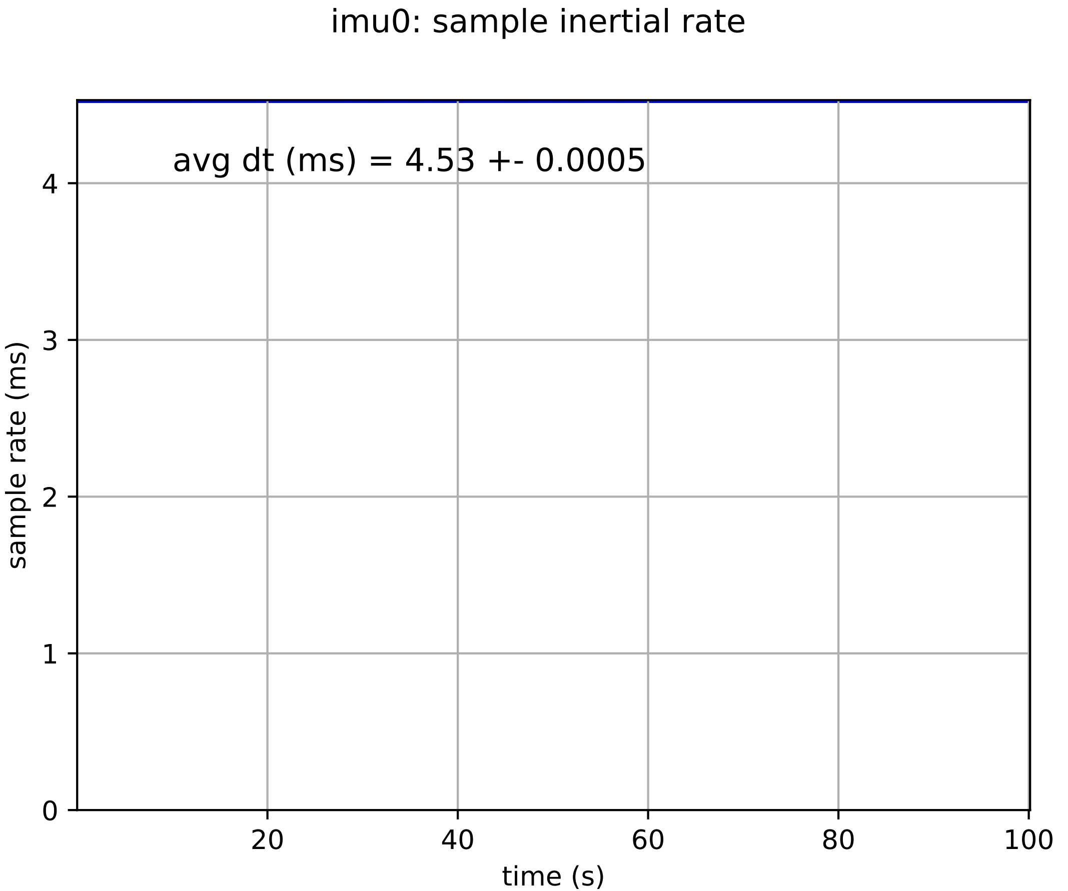
IMU0:

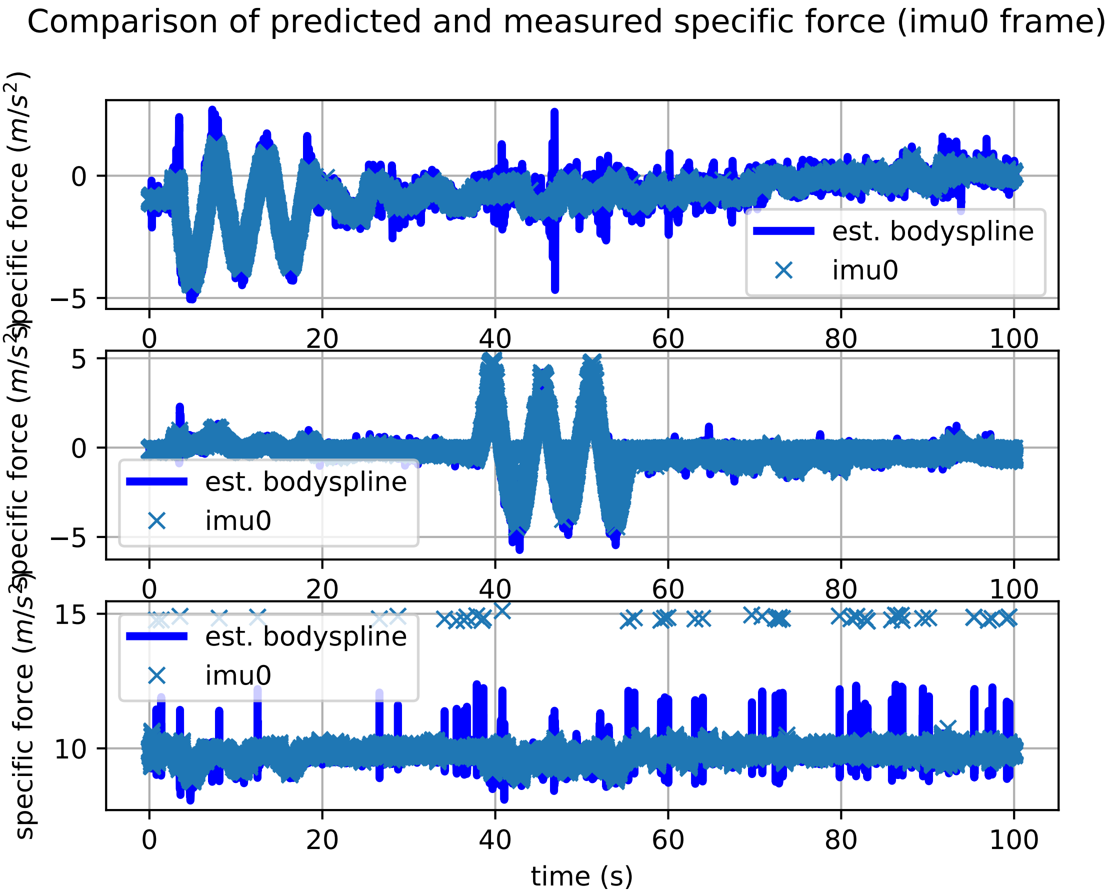
Model: calibrated
Update rate: 208.0

Accelerometer:
Noise density: 0.2330758294
Noise density (discrete): 3.3614674158919886
Random walk: 0.01339530913
Gyroscope:
Noise density: 0.5950319985
Noise density (discrete): 8.581673524534246
Random walk: 0.02296163583
T_ib (imu0 to imu0)
[[1. 0. 0. 0.]
[0. 1. 0. 0.]
[0. 0. 1. 0.]
[0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

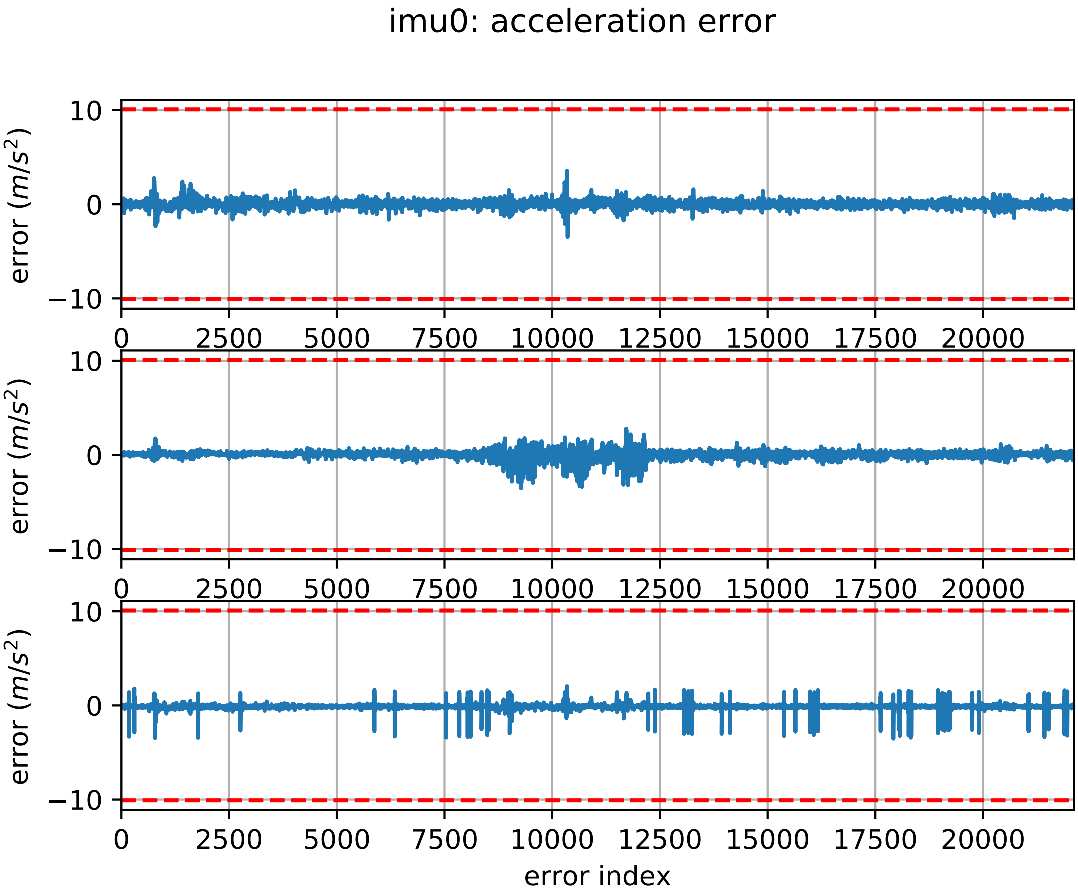
imu0: estimated poses

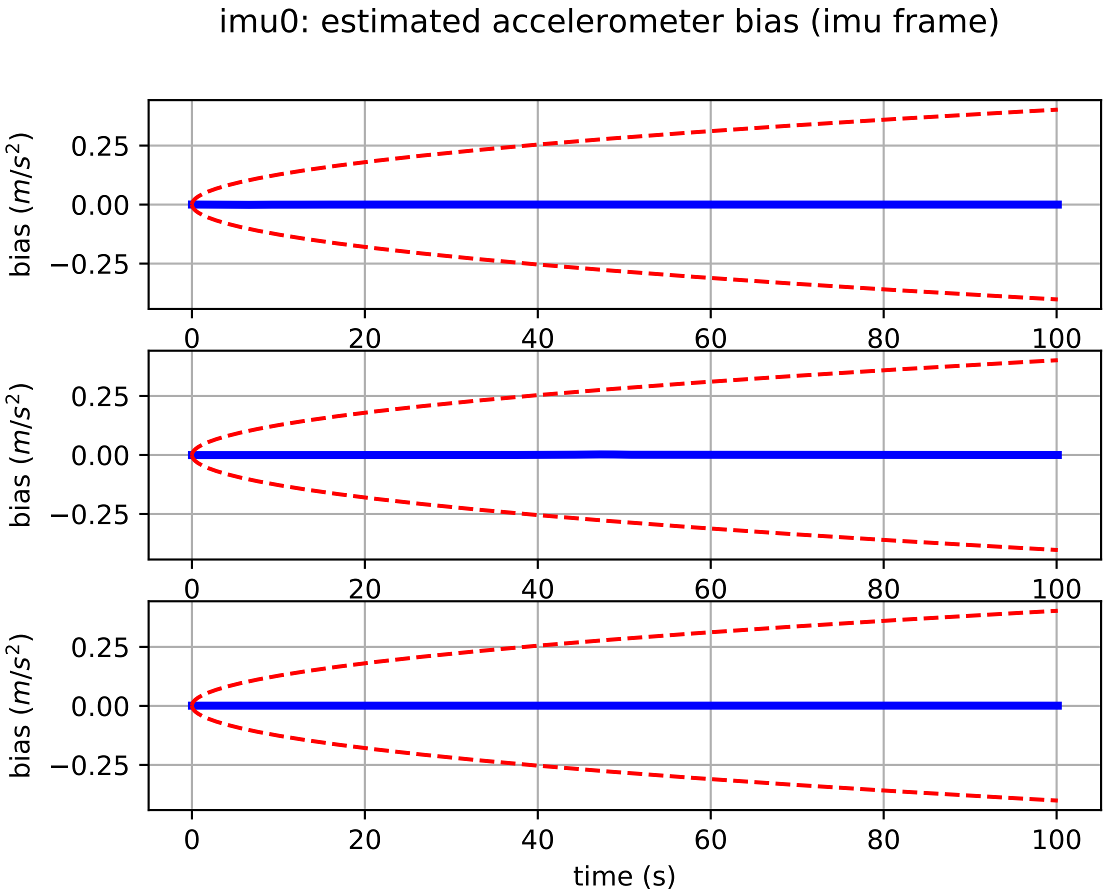


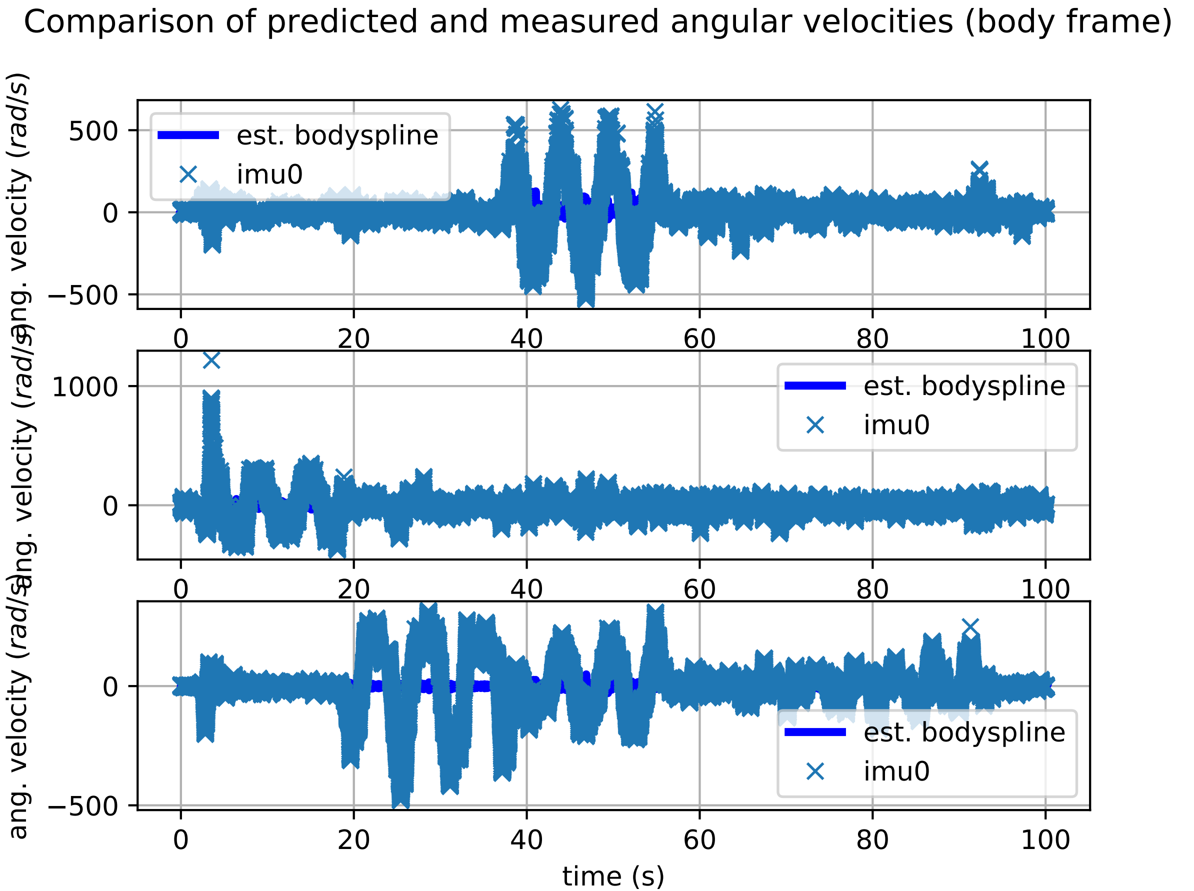

imu0: sample inertial rate
avg dt (ms) = 4.53 +- 0.0005
sample rate (ms)
time (s)
0
1
2
3
4
20
40
60
80
100

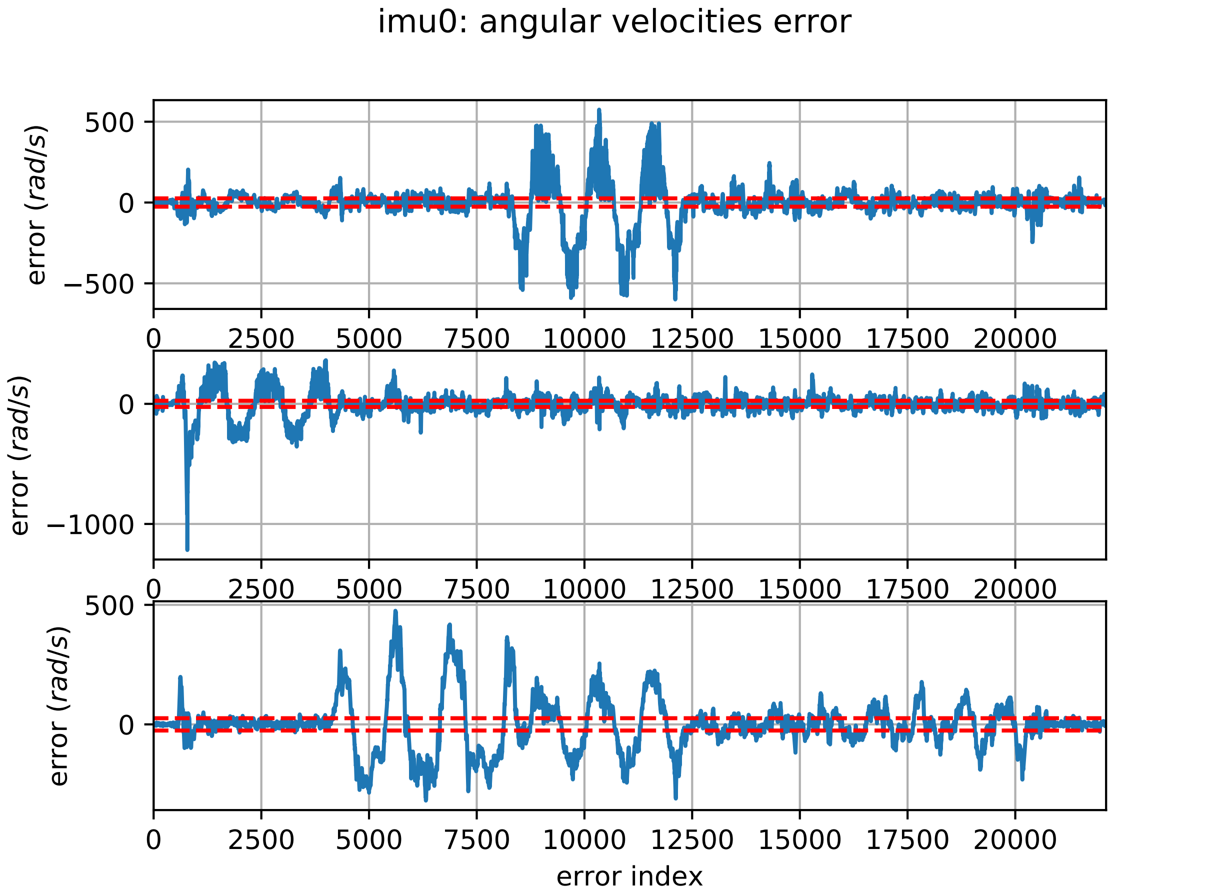


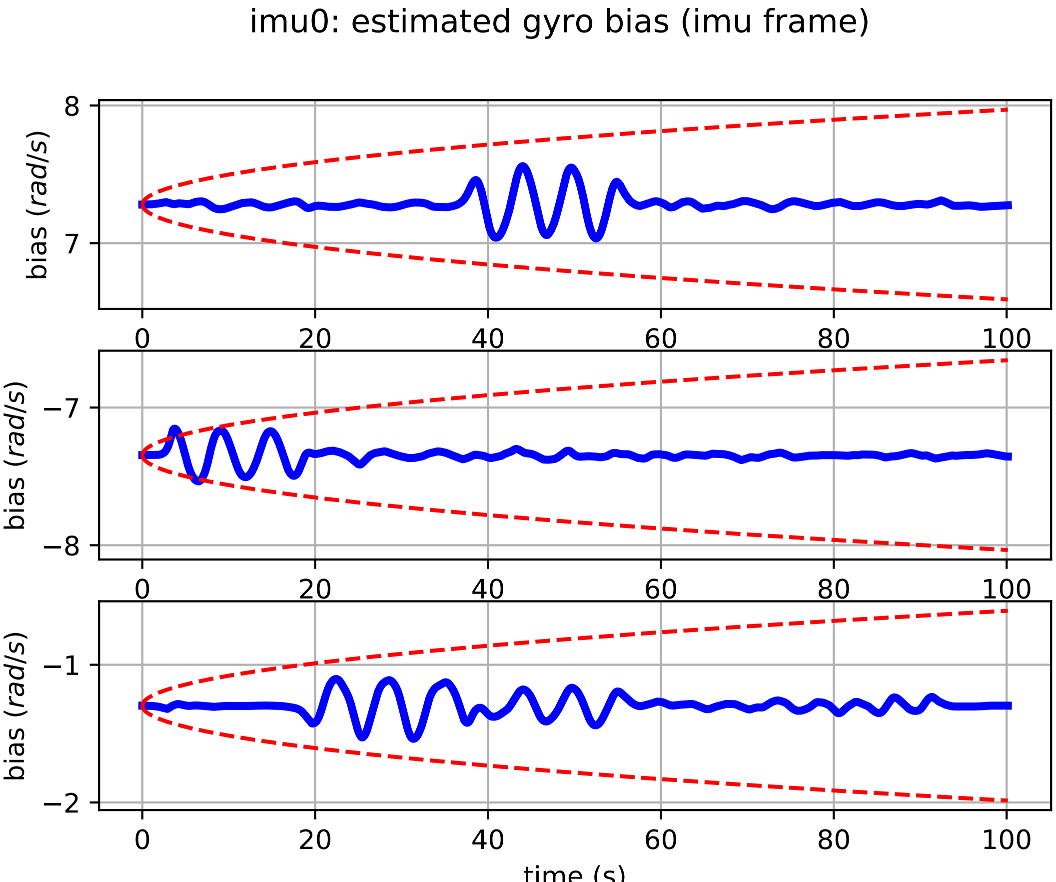
Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
est. bodyspline
imu0
0
−5
5
15
10
0
20
40
60
80
100
time (s)


imu0: acceleration error
10
0
−10
error (m/s^2)
0
2500
5000
7500
10000
12500
15000
17500
20000
error index

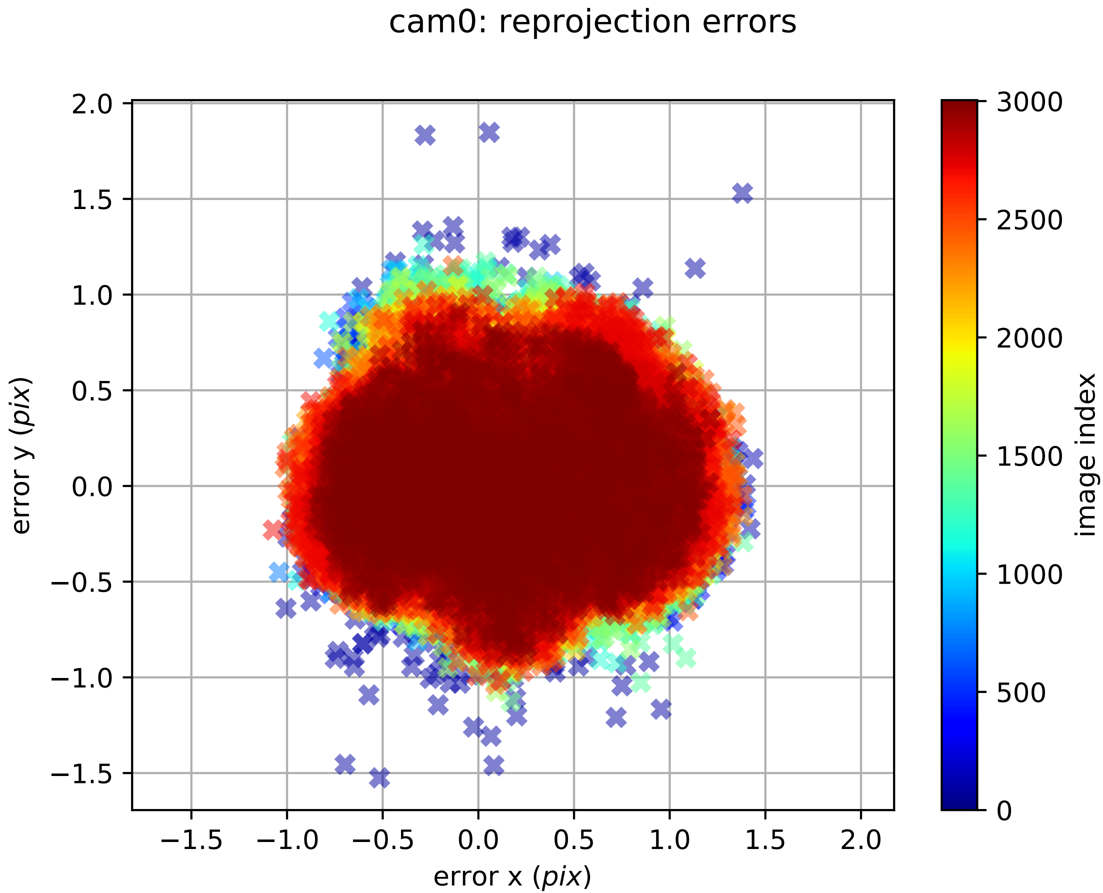

imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.25
0.00
−0.25
0
20
40
60
80
100
time (s)

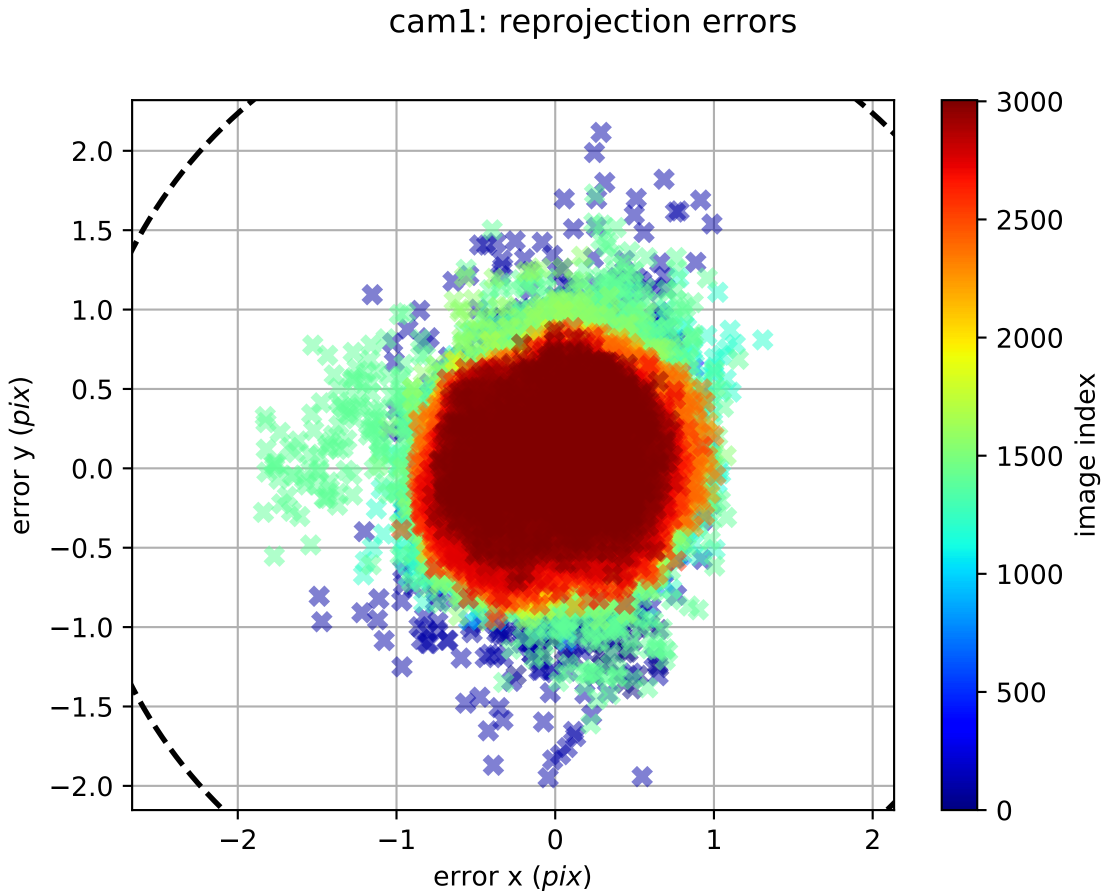

Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
est. bodyspline
imu0
500
0
−500
1000
0
20
40
60
80
100
time (s)


imu0: angular velocities error
error (rad/s)
500
0
−500
error (rad/s)
0
−1000
error (rad/s)
500
0
0
2500
5000
7500
10000
12500
15000
17500
20000
error index



imu0: estimated gyro bias (imu frame)
bias (rad/s)
8
7
−7
−8
−1
−2
0
20
40
60
80
100
time (s)


cam0: reprojection errors
error x (pix)
error y (pix)
image index
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500
2000
2500
3000



cam1: reprojection errors
error y (pix)
error x (pix)
image index
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2
−1
0
1
2
3000
2500
2000
1500
1000
500
0